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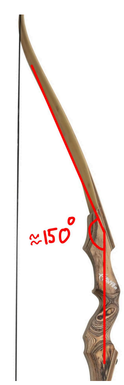
Autumn 2025

ME 80 Final Project Report - Archery Recurve Bow Analysis

Introduction: A section introducing your project and why you chose it.

During the initial brainstorming stages of the project, while discussing project ideas, we came across the idea of analyzing a bow and arrow. As we compared ideas, we realized we all shared an interest in archery—two of us planning to join the Stanford Archery Club, while the other had prior archery experience. With the general idea of a bow and arrow decided, we began to dive into questions. While Googling bows, we were shocked to see the number of distinct shapes. We noticed how many distinct categories of bows existed (training bows, recurves, hunting bows, and more), and how each category tends to have its own consistent "standard" shape, thickness profile, and overall geometry. That consistency made the designs feel intentional rather than aesthetic: recurves, in particular, often share a tapered profile with more material near the grip and less toward the limb tips, and we wanted to understand what functional role that taper actually plays. This led us to focus our project on analyzing the structural design and materials of a recurve bow, with a specific emphasis on why recurve limbs taper off and whether that geometry is truly the most power-efficient way to store and transfer energy during the

Design: A section describing the overall design concept along with the rationale behind major design decisions. This section should also include images of the CAD models.



We wanted to implement the following concepts from the course: stress & strain, beam bending, and material selection. To best use the skills we learned from the course, we decided to simplify our analysis. We are analyzing a bow separated into three sections: 1 handle rod and two "arms". The handle rod connects the two arms at a 150-degree angle. We decided on 150 degrees by analyzing images of recurve bows. Upon further calculations, we realized that this simplified version of the bow of having our bow be sectioned into 3 rods, is more accurate to an actual bow than a single curved rod. Actual recurve bows are designed with a stronger metal center for the handle, which does not bend, but serves as support for the arms as they bend to store the energy from pulling it back.

Through our research, we learned that the average youth/teen bow length[1] ranges from 48 inches to 62 inches, with a draw weight between 10-35 pounds. For our simulated life-size bow, we decided to have the arms of our bow be 24 inches, the handle 4 inches, and the draw weight to be 20 pounds. It was also noted that the thickness of the recurve bows is typically between

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2 Standard dimensions of recurve/long limbs at the top/bottom of the Riser?. saddlehunter.com. https://saddlehunter.com/community/index.php?threads%2Fstandard-dimensions-of-recurve-long-limbs-at-the-top-bottom-of-the-riser.40519%2F

0.25 inches and 0.75 inches, so we decided to assign our bow's thickness to be 0.5 inches[2]. For the designs, we created six different scaled versions of this simulated life-size bow, at a ratio of 1:4, which resulted in the following measurements: Arms – 6 inches, Handle – 1 inch, Thickness – 0.125 inches, Draw weight – 5 pounds.

We grouped the bows into three design categories, each of which includes both versions (rectangular arms and tapered arms): straight rod, curved rod, and filleted curved rod. Our project investigates whether the curved, rounded, and tapered arms commonly found in commercial recurve bows truly offer the best performance in terms of stress distribution and deformation. We evaluate this by analyzing a range of bow shapes and design variations. Our hand calculations are for the tapered and angled design. The designs and FEA simulations were done under the assumption that the force would be applied at a 90-degree angle from the ends of the legs for the angled designs and at a 45-degree angle for the straight designs.

Analytical Results: A section describing the setup and assumptions for your analytical calculations. This section should NOT walk through the calculations line by line (you should include the calculations somewhere; perhaps in an appendix). Instead, it should provide an overview of how you approached the calculation and what relevant details someone trying to reproduce your calculations would need to know.

To analyze our scenario, we used beam bending techniques followed by material selection with mass optimization. Both of the arms can be simplified to cantilevered beams. The stress of a cantilever beam is defined as the following:

$$\sigma = \frac{M(x)y(x)}{I(x)}$$

This equation is for a variable cross-section with x defined as the distance between one end of the beam to another. Because the cross-section of our beam is a rectangle and we are keeping the thickness the same, we can write y(x) and I(x) in terms of b(x) or how the base changes in terms of the distance from the end of the beam that is attached to the central beam. These equations can be substituted into our original stress equation to get the following:

$$\sigma = \frac{6T\sin(a)(L-x)}{t^2 b(x)}$$

And by extension:

$$b(x) = \frac{6T\sin(a)(L-x)}{\sigma t^2}$$

This equation represents a linear relationship between how large the base needs to be to accommodate stress at a certain point in the bow makes sense because for all of our relationships with inertia and moment, the base of the cross section and our x variable. Using our base relationship, we can also find equations for mass and displacement.

$$\delta(x) = \frac{4L^3\sigma}{Et(L-x)}$$

$$m(x) = \frac{6\rho LT\sin(a)(L-x)}{\sigma t^2}$$

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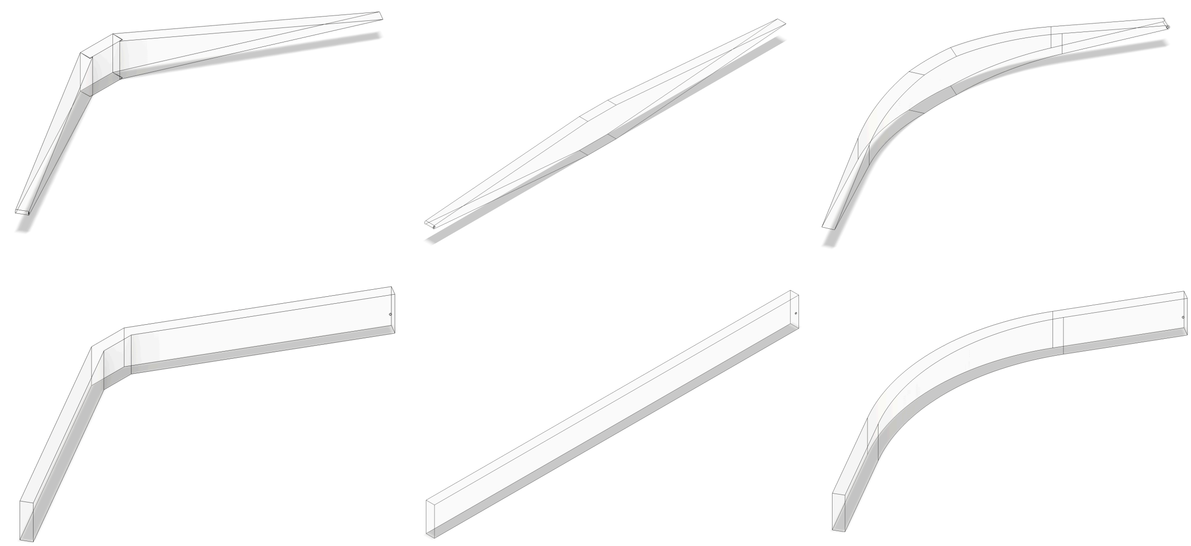
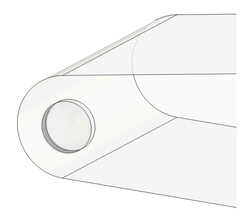
Simulation Results: A section describing the setup of the FEA simulation with enough detail that someone else could reasonably reproduce your work. This section should also include the major results of your simulation along with images of the simulation. You may include other interesting results in an appendix.



Left to right: row 1 – Taper Angled, Taper Straight, Taper Fillet; row 2 – Rectangular Angles, Rectangular Straight, Rectangular Fillet



To most accurately simulate drawing an actual bow in our FEA simulation, we decided to use the design's symmetry to our advantage and run the simulation on only the top half of the bow, and fix where this half would connect to the bottom half. Initially, we constrained the outer face of the bow, did not fillet the ends of the tapered bows, and applied the force to the inner corner of the bow. After we ran this initial simulation, we realized that this applied a shear force, and it did not distribute the force throughout the bow. We also noticed that fixing the outer face induced forces and moments to keep the face of the bow from moving, which doesn't reflect the bow's true loading conditions. Additionally, we made a small divot near the end of the bow to simulate where the string would pull the ends of the bow. For the tapered bows, the tips of the bows were filleted with a 0.05-inch diameter. We added a 0.05-inch diameter hole 0.025 inches from the edge, and we negatively extruded this circle 0.01 inches into the bow. We used this indented surface to apply 5-pound forces perpendicular to the angled arms and at a 45-degree angle for the straight arms. For the simulation, we assigned our material to be clear acrylic because it had a yield stress of 40 MPa, which is the same as PLA.

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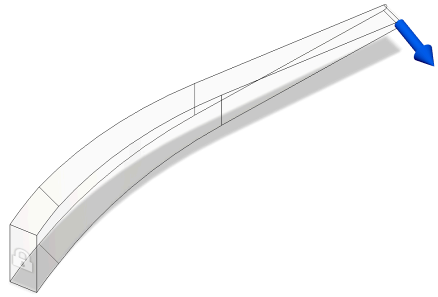
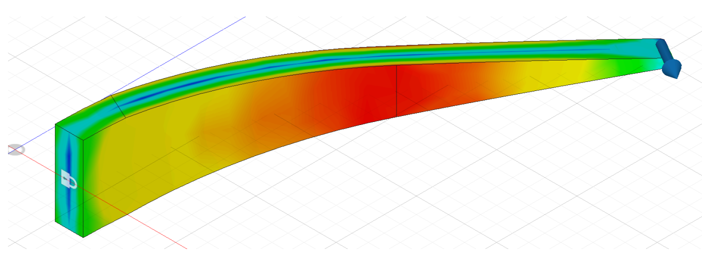


Set up of how we did FEA testing.

	Max. Von Mises (MPa)	Min. Factor of Safety
Taper Angled	16.315	2.452
Taper Straight	19.273	2.07
Taper Fillet	**10.19**	**3.953**
Rectangular Angled	12.437	3.216
Rectangular Straight	11.161	3.854
Rectangular Fillet	**9.329**	**4.288**

We noticed that for both the tapered and rectangular designs, the design that was filleted had the highest minimum factor of safety while having the maximum Von Mises stress. This aligned with what we expected from the designs. With the rectangular fillet having a higher minimum factor of safety and a lower maximum Von Mises stress, it would be natural to assume that the rectangular fillet is a better design for the bow. However, the rectangular fillet design is double the weight of the taper fillet. For half of the weight, the taper fillet design was able to perform better than almost all of the designs.



One notable outcome was that the Taper Angle and Taper Straight designs yielded a factor of safety below our target threshold of 3. We hypothesize that this could be because of the loading conditions and geometry, particularly the filleted ends and the placement of force on the small cut hole instead of the inner corner or edge. The current factor of safety values are lower

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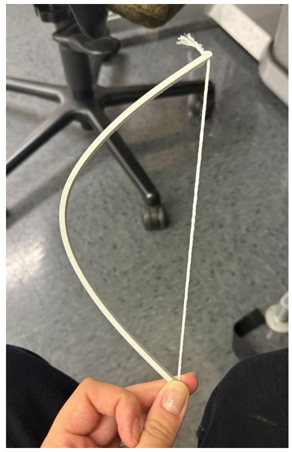
than our hand calculations because those calculations were based on an idealized model that assumed a pointed bow tip and the load applied at the very end, which simplifies the stress profile. The FEA, however, accurately captures the stress applied to an actual bow because, realistically, the string of a bow cannot physically be strung to the extreme edge of the bow. Our divot imitates that of a real bow by placing the force slightly inside from the end.



In our prototype, we simply printed a single beam and drew it taut with a piece of string. While it was incredibly simple, it showed us what we needed to know about stresses distributed in the bow. Most commonly, recurve bows have two arms and a thicker or stronger handle at the center. While this can seem just to be for the comfort of shooting, it seems it also has another, more important purpose of bracing the bow at maximum draw so that it doesn't fail. In our initial prototype, it was soon obvious that the beam was under the maximum stress at the center, where we could see the material splitting.

Discussion: This section should discuss any interesting findings from either the calculations or simulations, along with a comparison of any discrepancies between the two and potential causes for the discrepancies. You should also discuss any challenges you had in making your prototype and any differences between it and the CAD design.

One of the first things we noticed was the variety of factor of safety values that were produced from the FEA testing. From our calculations, we assumed a value of 3 following industry standards, and we expected the FEA testing to result in factor of safety values of around 3. We believe it is because of the placement of the force for the simulation. Our calculations for a 6-inch arm with a load applied at the very end, but in the FEA, as described in the simulation section, having a load on a tapered edge, which essentially goes to an infinitely small edge, created a singularity and caused the FEA to give us inaccurate results. To make our CADs and

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FEA more accurate to our prototypes, we rounded the sharp edge on the edge and applied the force slightly inward from the very edge, which accounts for the difference.

The main difference between our CAD design and our prototype is that we simplified the string into two loads on either end of the bow for our CAD and FEA designs. For our prototype, we had an actual string and determined the loads by pulling the string back to a given distance. When the string is pulled, the bow deforms as expected, but there is still some complication in simplifying the tension caused by a string to two loads applied on either end of the bow.

Final Product: Include pictures of your final product in this section and accompanying descriptions for the pictures.

For our final product, we 3D-printed the Taper Angled design and tied a string to the ends of the legs to simulate the forces implemented in an actual bow. It was printed with PLA at 100% infill, following the conditions used for our FEA testing. For testing, we realized that it would be difficult to accurately imitate the FEA testing without a flat center panel, which cancelled out the filleted and straight designs for the bows. Between the tapered and rectangular, we wanted to experiment with the most realistic bow, so we decided to print out the tapered design.

To simulate the applied forces in the FEA and also in real life, we held the center panel in place while applying 5 pounds of force to the center of the string in the negative x direction. Without a way to get the stress in the center of the bow, we relied on deformation or displacement of the ends of the legs to simulate the testing that we did. We noticed that the ends of the legs barely moved in the simulation, and when applying 5 pounds of force, we also noticed little to no displacement.


Conclusion: This section should tie all of the previous sections together including details about significant overall findings, interesting lessons learned, and future work. You should also discuss if your design met your initial goals and how it can be improved if it didn't or how you can be more sure of its structural integrity if it did.

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The goal for our project was to analyze and prove that the current standard bow is the most optimized design for a recurve bow. In other words, we wanted to analyze why recurve bows are built the way they are. Through FEA testing, beam bending calculations, and force calculations, we were able to confirm why a tapered filleted bow is the current standard for recurve bows. Through experimentation, we realized that rather than the tapered fillet bow being significantly *more* stable, it was that it was able to provide the *same* amount of support despite being half of the weight. Another interesting finding was the impact of the design being filleted versus having edges.

For future work, we could vary the variables that were kept constant in this experiment. Some variables we could vary include the thickness of the bow, changing the shape of the handle, having a more/less extreme angle for the legs, and filleting the entire design. We briefly analyzed a bow with fillets between the center and arm beams, which increased the minimum factor of safety. We assume that this is because sharp angles make the design more unstable; however, it would be interesting to provide a full analysis for this change. Another feature that we did not vary is the design of the handle. As the pivot point of the bow, it would be interesting to analyze the impact of varied designs of the handle of the bow.

Peer Review: During the project, team members should give constructive feedback to other members about their work. For example, the FEA and making person should provide feedback on the initial CAD design that can include critiques specific to their role i.e. the making person may think that the design isn't manufacturable or the FEA person may think a certain aspect may make it hard to simulate reliably. This should be done for each role by the two people not in that role. For the report, you should include this written feedback in a series of short paragraphs, with a sentence or two about how the feedback was incorporated into the final product.

Luca & Madison's feedback for Designer (Ellen): When discussing our design, we quickly realized the importance of standardizing the naming conventions for dimensions and components. We defined what "base," "thickness," and "width" meant, which cleared up a lot of confusion about what we were varying and what would remain constant. We also realized that the initial measurements that we decided on might not be practical, and changed accordingly. For instance, we decided on having the bow's thickness to be ¼ inches, but we soon realized that that would be way too thin. Finally, we changed the CAD to create new bodies with extrusions instead of joining them so the FEA person could simulate the different forces on each part of the bow. All of these changes were made in our final CAD.

Luca & Ellen's feedback for FEA Analyst (Madison):
For the FEA testing, the forces were placed on the very ends of the bows, which is different from where the tension will be for an actual bow. An actual bow would have a hole for the string about an inch from the end of the arms. Although these differences are slight, the force being applied on the very ends versus an inch into the arms results in different stresses and analyses.

Ellen & Madison's feedback for Manufacturer (Luca):
With a straight beam design, the structure lacks support in the middle when bent with a string. The moment varies throughout the beam, with the maximum stress acting in the middle. We

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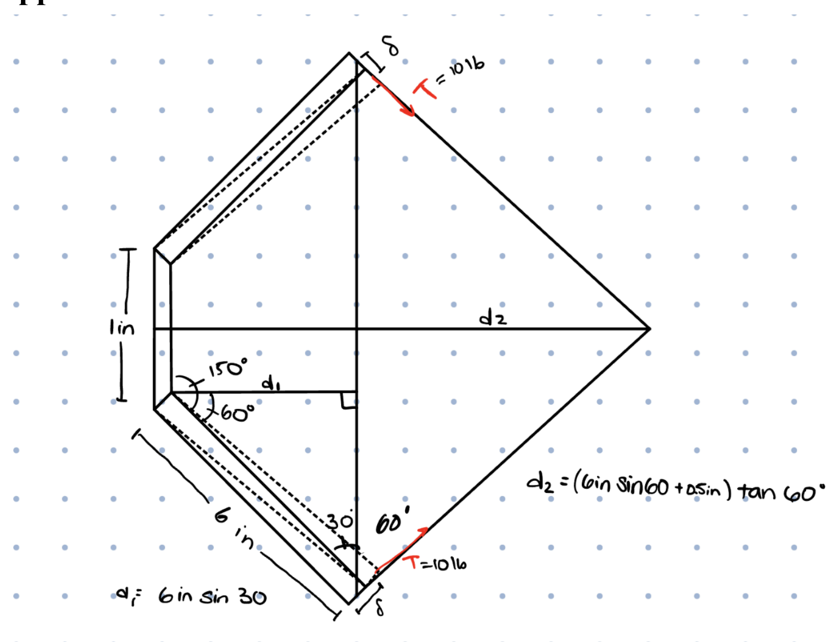
could either decrease the tension or enforce the center somehow, where the stress is greatest. It is also difficult to use FEA loads to properly mimic the force of tension of a string attached to both ends of the bow, as manufactured with the initial prototype. FEA, as we have learned in class, doesn't allow for a tension of one string attached to both ends with an initial tension. Because of this feedback, we decided to make a pre-curved bow where the angle could be determined and acted in a way that would make sense, rather than relying on the straight beam deforming to a curved structure. The moment is the same throughout the middle section and is our main constraint, which is reflected in our equations.

Appendix:



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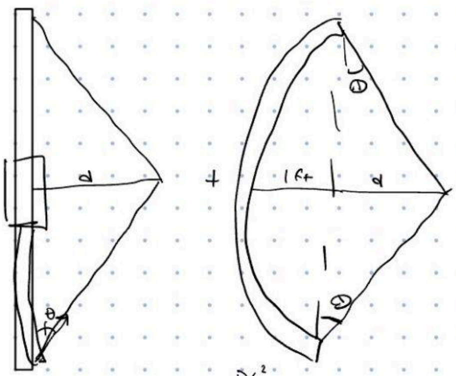
$L_1 = \frac{1}{6}\pi r 2 \qquad L_2 = \frac{1}{2}\pi r 2$

$L_1 = \frac{1}{3}\pi r \qquad L_2 = \pi r$

$V_1 = \frac{-T_1 \sin\theta_1 (\frac{1}{3}\pi r)^3}{3E(\frac{bh^3}{12})} \qquad V_2 = -\frac{T_2 \sin\theta_2 (\pi r)^3}{3E(\frac{bh^3}{12})}$



Looking for the relationship between length of bow arms and pondage (T)

$T_1 = T_2$

$V_1 = -T_1 \frac{4\sin\theta_1 \pi^3 (14in)^3}{27 E bh^3} \qquad V_2 = -\frac{4 T_2 \sin\theta_2 \pi^3 (14in)^3}{E bh^3}$

$-\frac{27 V_1 \cancel{Ebh^3}}{\cancel{4}\sin\theta_1 \cancel{\pi^3} \cancel{(14in)^3}} = \frac{-V_2 \cancel{Ebh^3}}{\cancel{4}\sin\theta_2 \cancel{\pi^3} \cancel{(14in)^3}}$

same material, same cross section

$\frac{27 V_1}{\sin\theta_1} = \frac{V_2}{\sin\theta_2}$

$\theta = \frac{PL^2}{2EI}$

$\delta = \frac{FL^3}{3EI} \Rightarrow F = \frac{3EI\delta}{L^3} \qquad I = \frac{bt^3}{12}$

$\sigma(x) = \frac{M(x)\,y(x)}{I(x)} \qquad \sigma_{max} = \frac{M\frac{t}{2}}{\frac{bt^3}{12}} \qquad m = \rho b t L$

$m(x) = \rho\, b(x) \cdot L t$

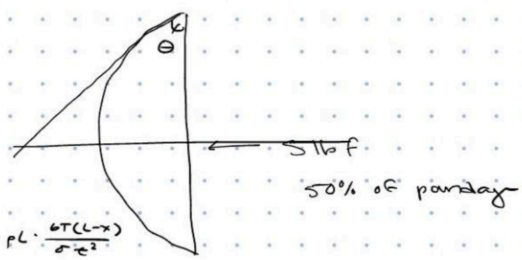
$\sigma_{max} = \frac{M(x)}{6 b(x) t^2}$

thickness same, base varies by x

$\sigma = \frac{T(L-x)}{6t^2 \cdot b(x)}$

$x = 0$

$m(x) = \rho L \cdot \frac{6T(L-x)}{\sigma t^2}$



$\delta = \frac{4TL^3}{t^3 E \cdot b(x)} \qquad b(x) = \frac{T(L-x)}{6\sigma t^2}$

$\delta = \frac{4TL^3}{t^3 E\, b(x)}$

$b(x) = \frac{4TL^3}{t^3 E \delta} = \frac{6T(L-x)}{\sigma \cancel{t^2}}$

$\delta = \frac{4TL^3}{t^3 E \cdot \frac{6T(L-x)}{\sigma t^2}} \qquad b_{max} = \frac{TL}{6\sigma t^2}$

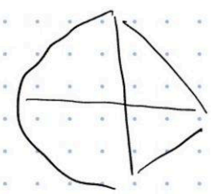
$\sigma = \frac{6(L-x) t E \delta}{4L^3}$

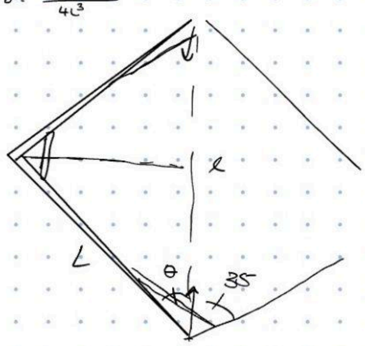
$\sigma = \frac{3}{2}\left(\frac{(L-x)tE\delta}{L^3}\right)$

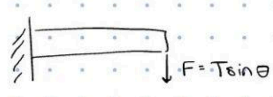
$\delta = \frac{4L^3\sigma}{tE(L-x)}$

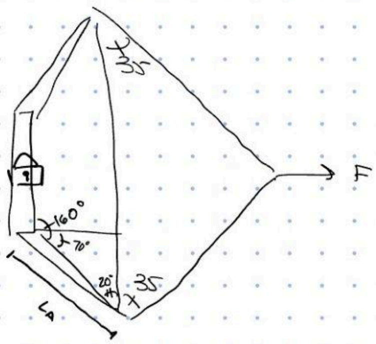
$\sigma_{max} = \frac{3}{2}\frac{tE\delta_{max}}{L^2}$

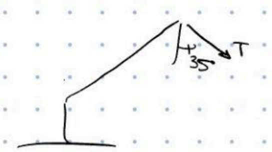
$\sigma = \frac{\delta t E(L-x)}{4L^3}$









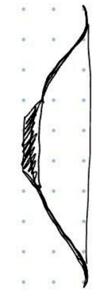


$\sigma_{max} = \frac{My}{I}$

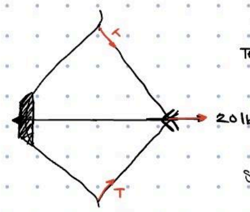
$= M(\frac{h}{2})(\frac{12}{bh^3})$

$= M\frac{6}{bh^2}$

+ fillet FEA



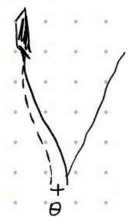
bows have different poundage
Ex: 20lb — takes 20lbF
to draw bow to full.



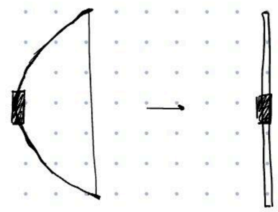
full draw: 28"

Tension when un-drawn: 0.3~0.5 of poundage
= 6lb ~ 10lb

Simplified to bending beam?



another question: Dry fire is really bad. Why?
google says: all energy is released back into bow
instead of into speed for arrow



Typical materials: $\underline{\text{Com Materials}}$
Maple Wood
Bamboo

rupture stress: Maple wood: 15800psi
Bamboo: 77Mpa ~ 107Mpa

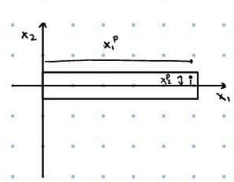
Strain energy of the bow:

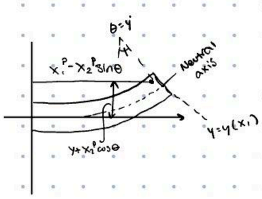
$SE = \frac{1}{2}kd^2$

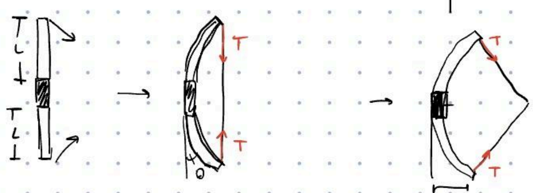
k - spring constant of bow

d - distance pulled

euler benoulli

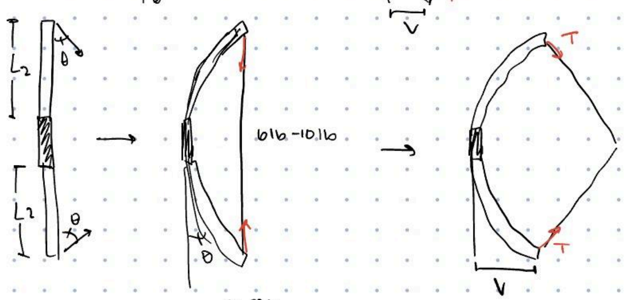


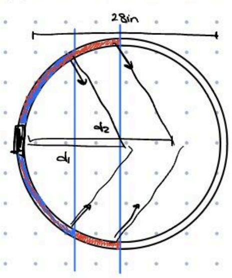




$$V(L) = -\frac{PL^2}{6EI}(2L) \qquad I = \frac{bh^3}{12}$$

$$= \frac{-T\sin\theta L^3}{3E\left(\frac{bh^3}{12}\right)}$$





1 *Beginner Guide to buying a bow: Pyramyd Air*. Pyramyd Adventures in Recreation. (n.d.). https://www.pyramydair.com/how-to-buy-bow
2 Standard dimensions of recurve/long limbs at the top/bottom of the Riser?. saddlehunter.com. https://saddlehunter.com/community/index.php?threads%2Fstandard-dimensions-of-recurve-long-limbs-at-the-top-bottom-of-the-riser.40519%2F